

```

#include <stdio.h>
#include <conio.h>
#define MAX 5
int front=-1,rear=-1;
int q[MAX];
void enq(int x)
{
    if(rear==MAX-1)
    {
        printf(" \n queue overflow");
    }
    else
    {
        if(front==-1)
        {
            front=0;
        }

        q[++rear]=x;
    }
}
void deq()
{
    if(front==-1 || front>rear)
    {
        printf("\nqueue underflow");
    }
    else
    {
        int item=q[front++];
        printf("\ndeleted item %d",item);
    }
}
void display(){
    if(front==-1)
    {
        printf("\n underflow");
    }
    else
    {
        printf("\nqueue elements:");
        for(int i=front;i<=rear;i++)
        {
            printf("%d ",q[i]);
        }
    }
}

int main()
{

```

```
int choice,x;
while(1){
printf("\n 1.Enqueue \n 2.Dequeue \n 3.Display \n 4.Exit");
printf("\n enter choice:");
scanf("%d",&choice);
switch(choice){

case 1:
printf("\n enter the element to be inserted:");
scanf("%d",&x);
enq(x);
break;

case 2:
deq();
break;

case 3:
display();
break;

case 4:
return 0;
break;

default:
printf("\n invalid choice");

}
}
return 0;
}
```